NTNU

Norwegian University of Science and Technology

OWASP - part I

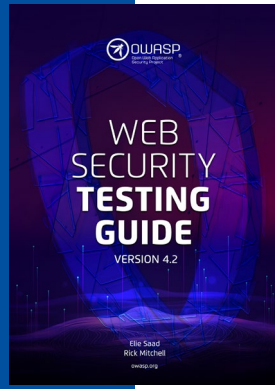
TDT4237 - 2026

Outline

Information gathering → Section 4.1

Injection attacks → Section 4.7

Session management attacks → Section 4.6



https://owasp.org/www-project-web-security-testing-guide/stable/

Reference group

Send an email to jingyue.li@ntnu.no by 6th of Feb.

Information gathering

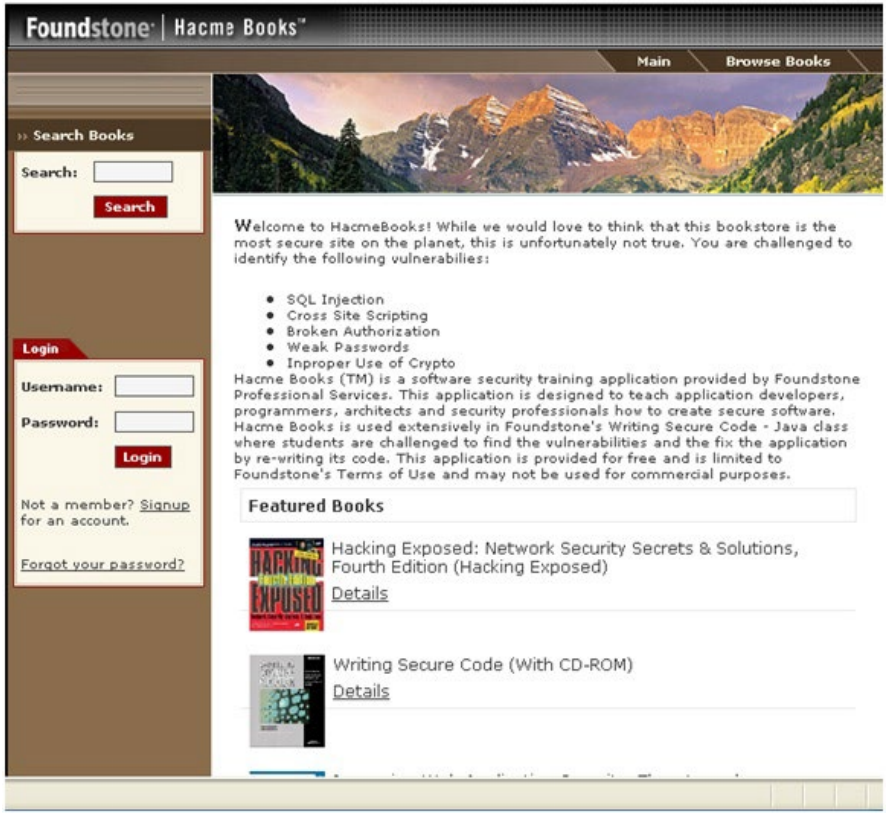
- Why information gathering?
  - Attacker
    - A map of points to attack
    - Look for low hanging fruit
    - Improve attack efficiency
  - Developer/internal tester
    - Decide test scope, coverage, prioritization
    - Improve test efficiency

The more you know about the application's structure, the better you can plan your tests!

What information to gather?

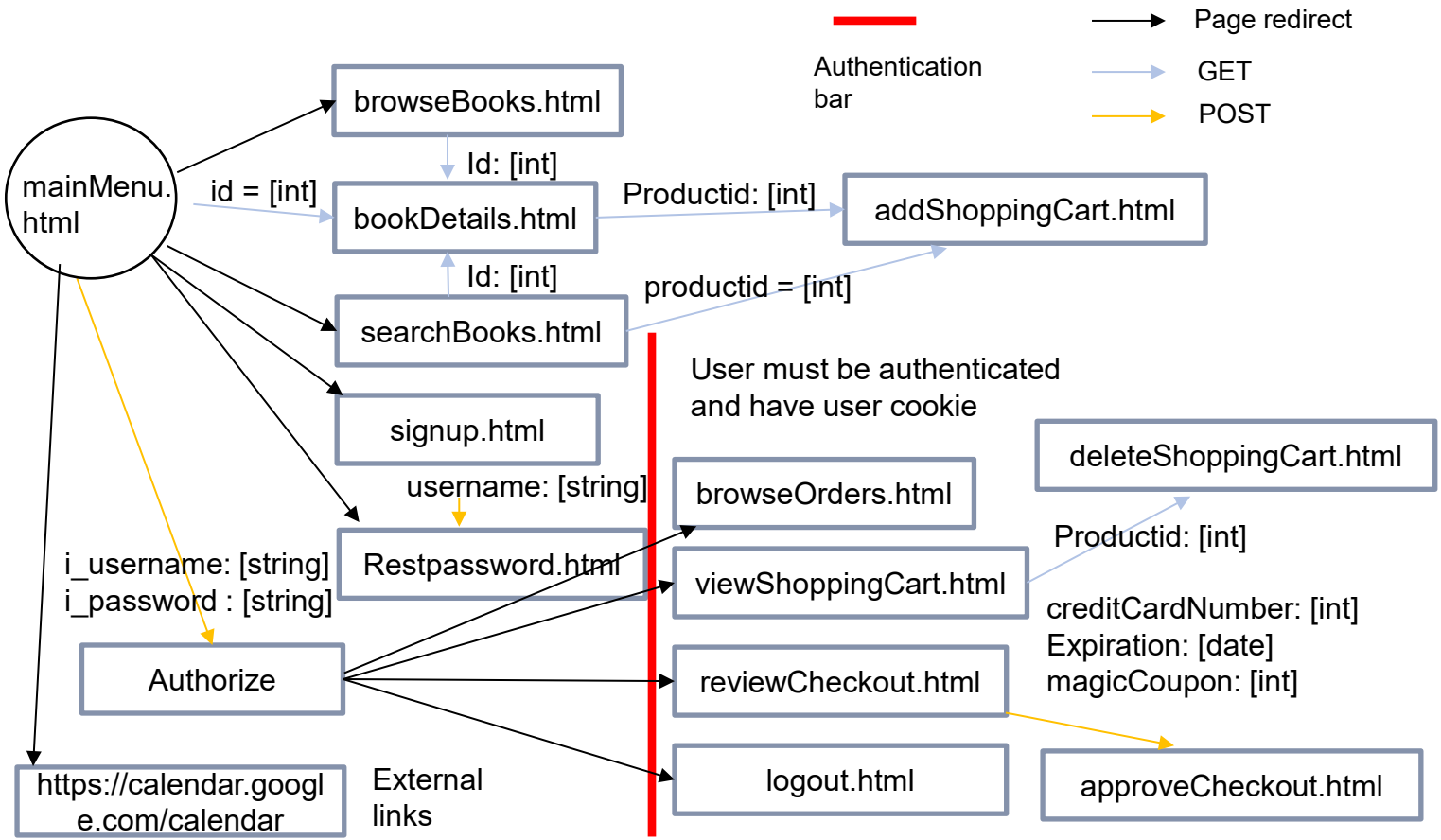
- Application structure
  - All pages you have found in the application
    - Including subdomains
  - Any external links
  - Trust zones
    - Needs authentication vs. open
- Data flow within the application, e.g.,
  - Parameters and values
  - Get and post, responses

Page map example - Hacmebooks



https://webapppentest.wordpress.com/2012/11/26/hacme-books-week-1/

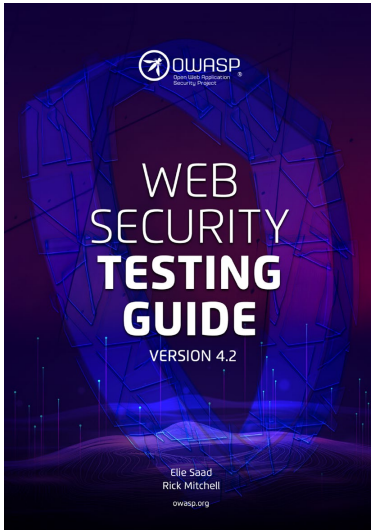
Simplified Hacmebooks page map



Other information to gather

- Infrastructure or platform, e.g.,
  - Web server (WSTG-INFO-02)
  - Applications on the webserver (WSTG-INFO-04)
  - Application entry points (WSTG-INFO-06)
  - Execution path through application (WSTG-INFO-07)
  - Web application framework (WSTG-INFO-08)

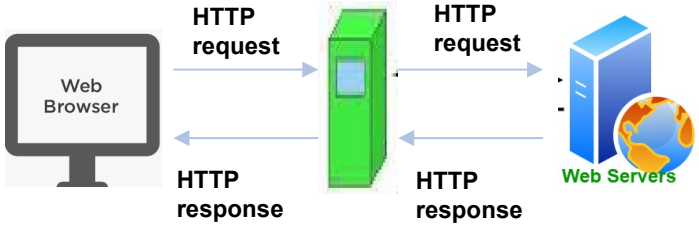
The IDs here refer to the ones in OWASP Web Security Testing Guide v4.2



Demo

Why use web debugging proxy?

- To capture and examine requests and responses
- To manipulate payloads
- Can also be used for attacks



Tools for information gathering

- Website copier (e.g., HTTtrack, VisualWget)
- Web debugging proxy server (e.g., Firefox Developer Tools, Fiddler)
- Tool sets (e.g., Kali Linux, Burp Suite and Zap)

HTTrack

Fiddler
Web Debugger
Loading AutoResponder...
v2.3.1.0

KALI LINUX

Burp Suite
Community Edition

ZAP

Automated scanners are limited

- Some information and vulnerabilities cannot be found using automated scanners
- Additional manual security testing is always recommended

Dorking (Google hacking)

EXPLOIT DATABASE

Google Hacking Database

Filters | Reset All

Show 15

Quick Search

Date Added	Dork	Category	Author
2022-09-19	intext:"index of" ".sql"	Files Containing Juicy Info	Gopalsamy Rajendran
2022-09-19	intitle:"index of" inurl:superadmin	Files Containing Juicy Info	Mahedi Hassan
2022-09-19	intitle:"WAMPSERVER Homepage"	Files Containing Juicy Info	HackerFrenzy
2022-09-19	inurl: json beautifier online	Files Containing Juicy Info	Nyein Chan Aung
2022-09-19	intitle:"IIS Windows Server"	Files Containing Juicy Info	HackerFrenzy
2022-09-19	intitle:"index of" inurl:SUID	Files Containing Juicy Info	Mahedi Hassan
2022-09-19	intitle:"index of" intext:"Apache/2.2.3"	Files Containing Juicy Info	Wagner Farias
2022-08-18	inurl:"index.php?page=news.php"	Advisories and Vulnerabilities	Omar Shash
2022-08-18	inurl:/sym404/root	Files Containing Juicy Info	Numen Blog
2022-08-17	inurl:viewer/live/index.html	Various Online Devices	Palvinder Singh Secuneus
2022-08-17	intitle:Index of "/venv"	Sensitive Directories	Abhishek Singh
2022-08-17	intitle:"WEB SERVICE" "wan" "lan" "alarm"	Pages Containing Login Portals	Heverin Hacker
2022-08-17	allintitle:"Log on to MACH-ProWeb"	Pages Containing Login Portals	Under The Sea hacker
2022-08-17	intitle:"index of"|"access_token.json"	Files Containing Juicy Info	Leonardo Venegas
2022-08-17	inurl:"admin/default.aspx"	Pages Containing Login Portals	Payal Yedhu

Showing 1 to 15 of 7,536 entries

FIRST PREVIOUS 1 2 3 4 5 ... 503 NEXT LAST

Downloads	Certifications	Training	Professional Services
Kali Linux	OSCP	Penetration Testing with Kali Linux (PWK) (PEN-200) All new for 2020	Penetration Testing
Kali NetHunter	OSWP	Offensive Security Wireless Attacks (WiFu) (PEN-210)	Advanced Attack Simulation
Kali Linux Revealed Book	OSEP	Evasion Techniques and Breaching Defences (PEN-300) All new for 2020	Application Security Assessment
	OSWE	Advanced Web Attacks and Exploitation (AWAE) (WEB-300) Updated for 2020	

https://www.exploit-db.com/google-hacking-database

Demo

Injection Attacks

Injection attacks

- SQL injection
- Blind SQL injection
- Xpath injection
- …

SQL injection – normal input

Username: Gandalf Password: TDT4237 Log In

"Server-side login code (E.g., PHP)"

$ result = mysql_query (" select * from Users where (name = '$user' and password = '$pass'); ");

Application constructs SQL query from parameter to DB, e.g.,

Select * from Users where name = Gandalf and password = TDT4237

SQL injection – Attack scenario (1)

- Attacker types in the string below in the ***username*** field

  Gandalf ’ OR 1=1); --

- At the server side, the code to be executed

$ result = mysql_query (“ select * from Users where (name = ‘Gandalf’ OR 1=1); -- and password = ‘whocares’); ”);

- SQL query constructed is

Select * from Users

where name = Gandalf OR 1=1

1=1 is always true.

SQL injection – Attack scenario (2)

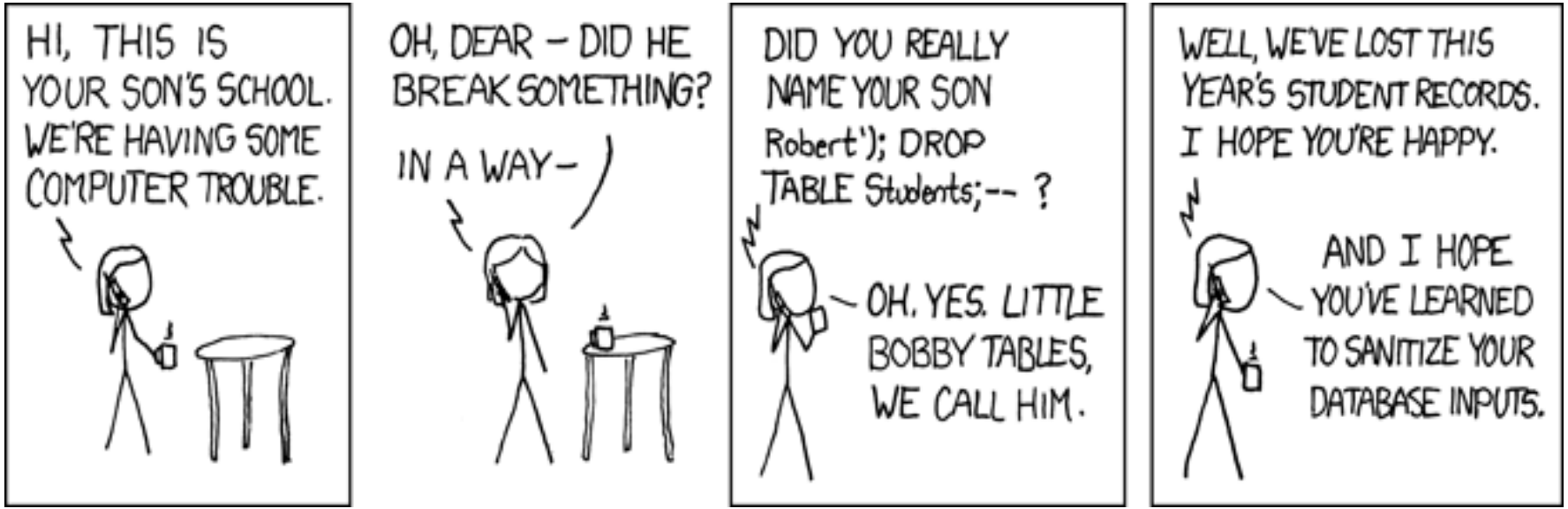
- Attacker types the following string in the ***username*** field

  Gandalf ' OR 1=1); Drop TABLE Users; --

- SQL query constructed is

```
select * from  Users
where name = Gandalf OR 1= 1;
drop TABLE Users;
```

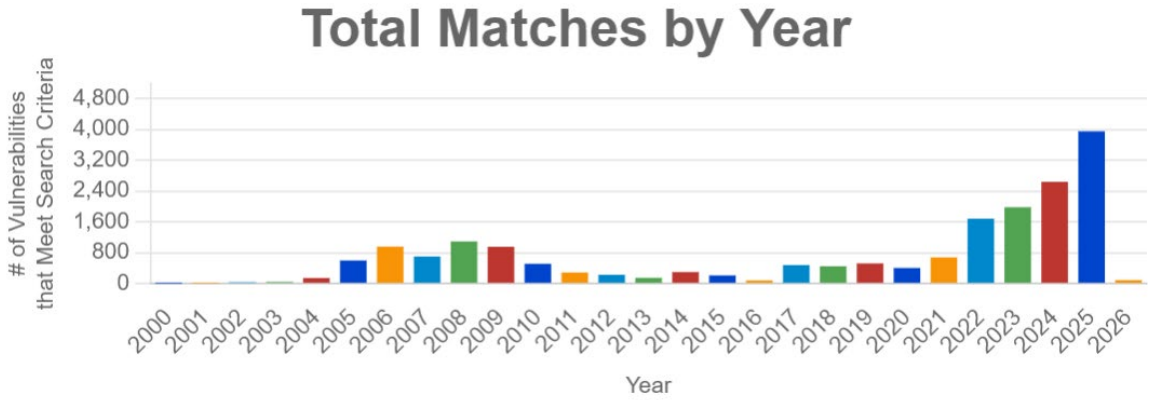
SQL injection humor

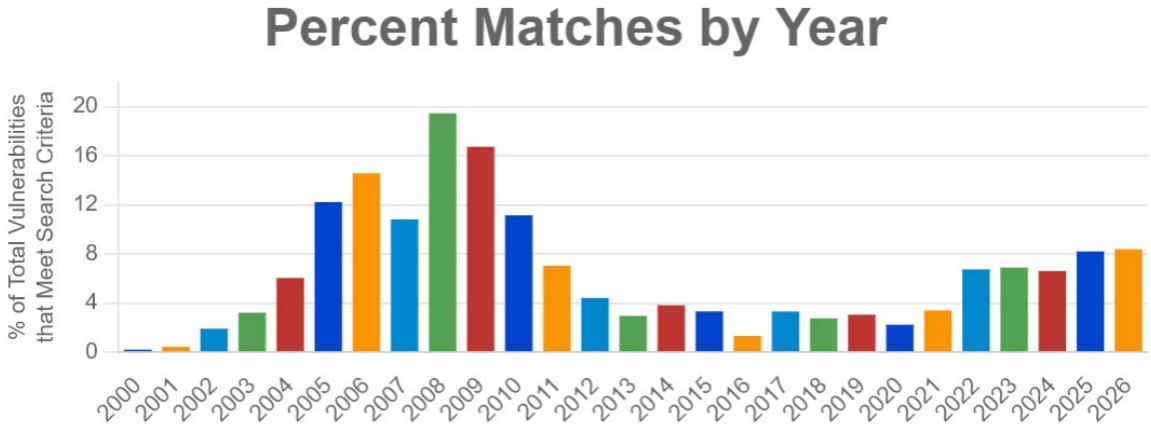


https://xkcd.com/327/

…not just humor

https://nvd.nist.gov/vuln/search





…some notable events

- **Tesla 2014:** Security researchers breached the website of Tesla using SQL injection, could gain administrative privileges and steal user data.
- **Fortnite 2019:** Fortnite is an online game with over 350 million users. A SQL injection vulnerability was discovered which could let attackers access user accounts.
- **WordPress 2022:** LearnPress plugin vulnerable, 75K sites impacted

https://brightsec.com/blog/sql-injection-attack/
https://www.indiehackers.com/post/sql-injection-real-life-attacks-and-how-it-hurts-business-c7ff42ef30

Why so common?

What can you achieve?

- Bypass authentication
- Privilege escalation
- Stealing information
- Destruction

Blind SQL injection

- Is the site vulnerable to SQL injection?
  - First register as a legal user, e.g. “Sauron”
  - Then, run SQL inject attack and see results

Sauron ’ AND 1=1); --

TRUE

Server side: SELECT Id FROM Users WHERE (‘userID= Sauron’AND 1=1); --

Info. Related to the Id shows → web app is vulnerable to SQL injection

Blind SQL injection (cont')

- Guess DB schema through a binary search

Q: What is the first letter of a Table in DB?

SELECT Id from Users WHERE userID= Sauron AND ascii(low (substring ((SELECT Top 1 name FROM sysobjects WHERE xtype = 'U'), 1, 1))) > 109

- First letter after m (ascii of m is 109), *"Id"* will show
- First letter before m, *"Id"* will not show

Xpath injection

User/password/account DB in XML (users.xml)

```
<?xml version=”1.0” encoding=”ISO-8859-1”?>
  <users>
    <user>
      <username>gandalf</username>
      <password>Abcd3</password>
      <account>admin</account>
    </user>
    <user>
      <username>Stefan0</username>
      <password>w1s3c</password>
      <account>guest</account>
    </user>
  </users>
```

Xpath injection (cont’)

- Normal Xpath query

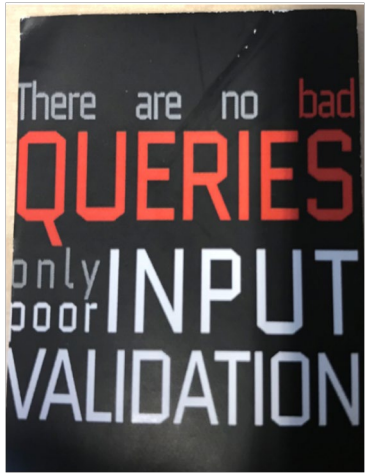
  string(//user[username/text()=’gandalf’ and password/text()=’Abcd3’]/account/text())

- Attack query

  string(//user[username/text()=’’ or ‘1’ = ‘1’ and password/text()=’’ or ‘1’ = ‘1’]/account/text())

SQL injection countermeasures

- Blacklisting
- Whitelisting
- Escaping
- Prepared statement & bind variables
- Mitigating impact



Blacklisting

Filter quotes, semicolons, whitespace, and …?

– E.g. kill_quotes (Java) removes single quotes

```
String kill_quotes(String str) {
  StringBuffer result = new    StringBuffer(str.length());
  for (int i = 0; i < str.length(); i++) {
    if (str.charAt(i) != '\'')
      result.append(str.charAt(i));
  }
  return result.toString();
}
```

user1 ' OR 1=1); --

Pitfalls of Blacklisting

- Could miss dangerous characters
- May conflict with functional requirements
  - E.g., a user with name O’Brien

Whitelisting

- Only allow well-defined safe inputs
- Using RegExp (regular expressions) match string
  - E.g., *month* parameter: non-negative integer
    - RegExp: ^[0-9]+$
    - ^ beginning of string, $ end of string
    - [0-9] + matches a digit, + specifies 1 or more
- Pitfalls: Hard to define RegExp for all safe values

Escaping

- Could escape quotes instead of blacklisting
  - E.g., Escape(O’Brien) = O’’Brien

INSERT INTO USERS(username, passwd) VALUES (‘O’’Brien’, ‘mypasswd’)

- Pitfalls: like blacklisting, could always miss a dangerous character

Prepared statements & Bind variables

- Root cause of SQL injection attack
  - Data interpreted as control, e.g., Gandalf ’ OR 1=1); --,
- Idea: decouple query statement and data input

Example of Java prepared statement

```
PreparedStatement stmt=con.prepareStatement("update emp set name=?
where id=?");

stmt.setString(1,“Gandalf"); //1 specifies the first parameter in the query

stmt.setInt(2,101);

int i=stmt.executeUpdate();
```

https://www.javatpoint.com/PreparedStatement-interface

Example of Python prepared statement

```
query = """Update employee set Salary = %s where id = %s"""

input = (8000, 101)

cursor.execute(query, input)
```

https://pynative.com/python-mysql-execute-parameterized-query-using-prepared-statement/

Mitigating impact

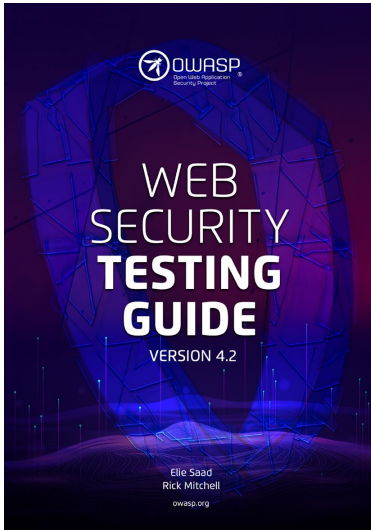
- Avoid information leakage
  - Don’t display a detailed error message to external users
  - Don’t display stack traces to external users
- Limiting privileges
  - No more privileges than users need
  - E.g., No drop table privilege for a typical user

Mitigate impact (cont’)

- Encrypt sensitive data, e.g.,
  - Username, credit card number, magical powers
- Key management precautions
  - Do not store the encryption key in DB
- Hash password

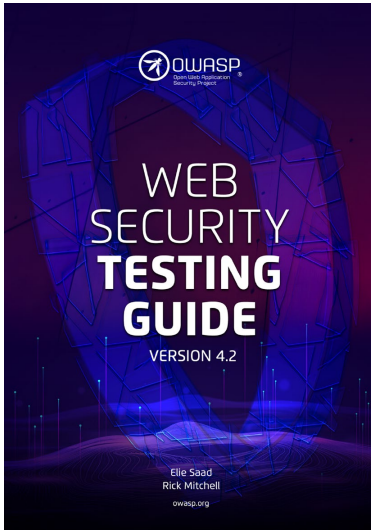
OWASP SQL injection test cases

- Testing for SQL Injection (WSTG-INPV-05)
  - Oracle Testing
  - MySQL Testing
  - SQL Server Testing
  - Testing PostgreSQL
  - MS Access Testing
  - Testing for NoSQL injection



OWASP other injection test cases

- Testing for LDAP Injection (WSTG-INPV-06)
- Testing for XML Injection (WSTG-INPV-07)
- Testing for SSI Injection (WSTG-INPV-08)
- Testing for XPath Injection (WSTG-INPV-09)
- IMAP/SMTP Injection (WSTG-INPV-10)
- Testing for Code Injection (WSTG-INPV-11)

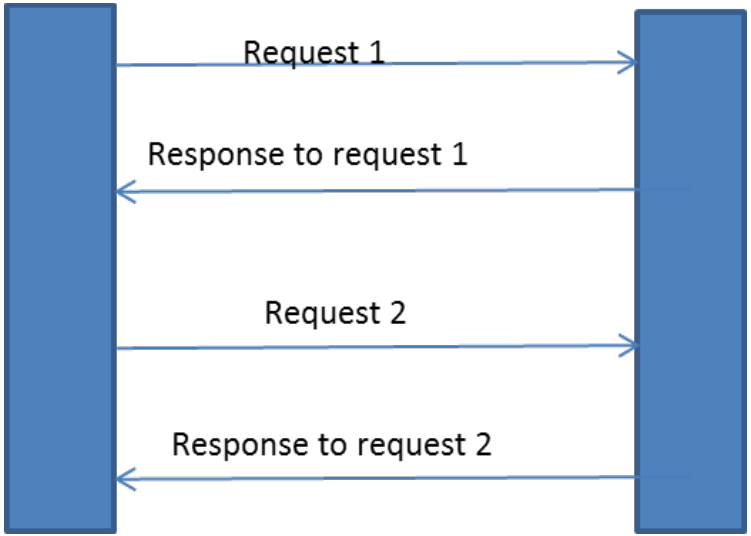


NTNU

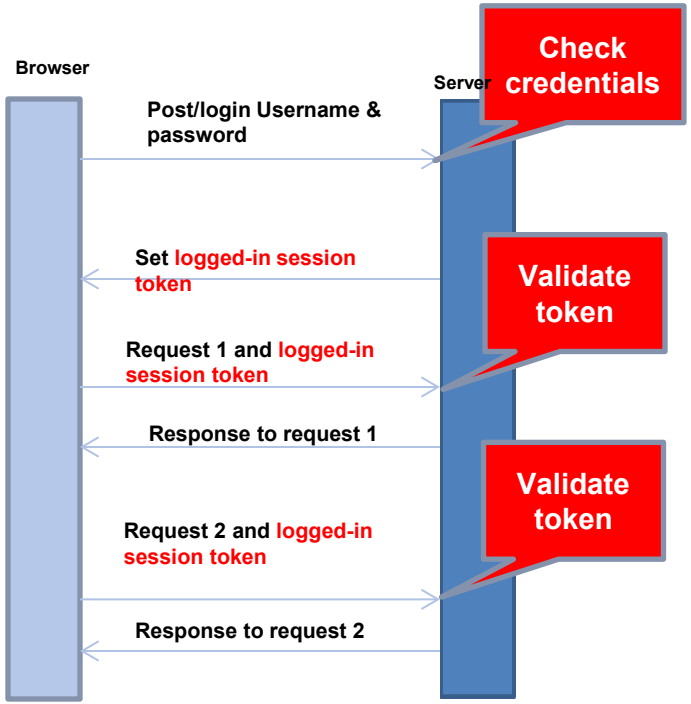
Session Management Attacks

Why session management?

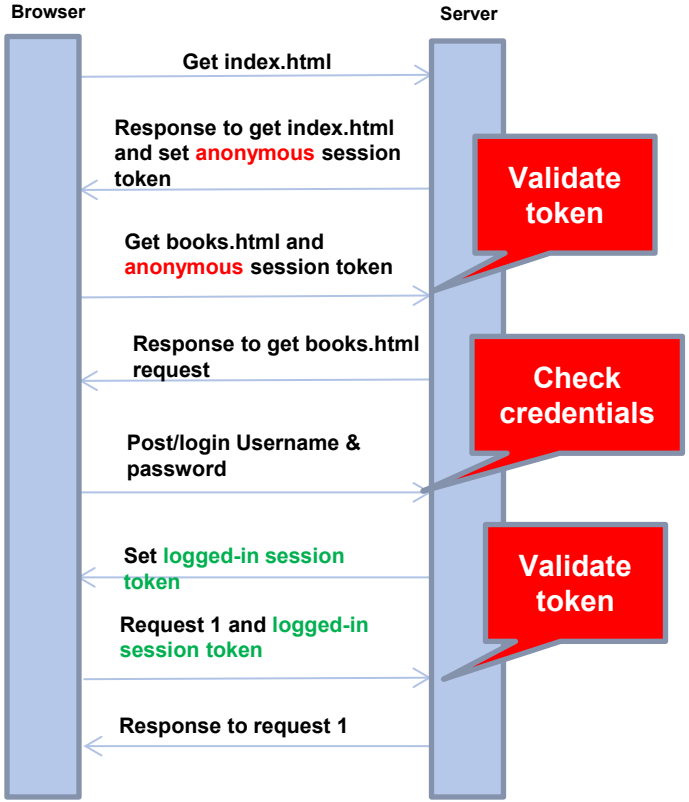
- HTTP is stateless
- Impossible to know if Req1 and Req2 are from the same client
- Users would have to constantly re-authenticate
- Session management
  - Authenticate user once
  - All subsequent requests are tied to the user



Session tokens



e.g., https://ntnu.inspera.no/admin



e.g., amazon.com

Where to store session token

- Embed in all URL links

  https://site.com/checkout?sessionToken= 1234

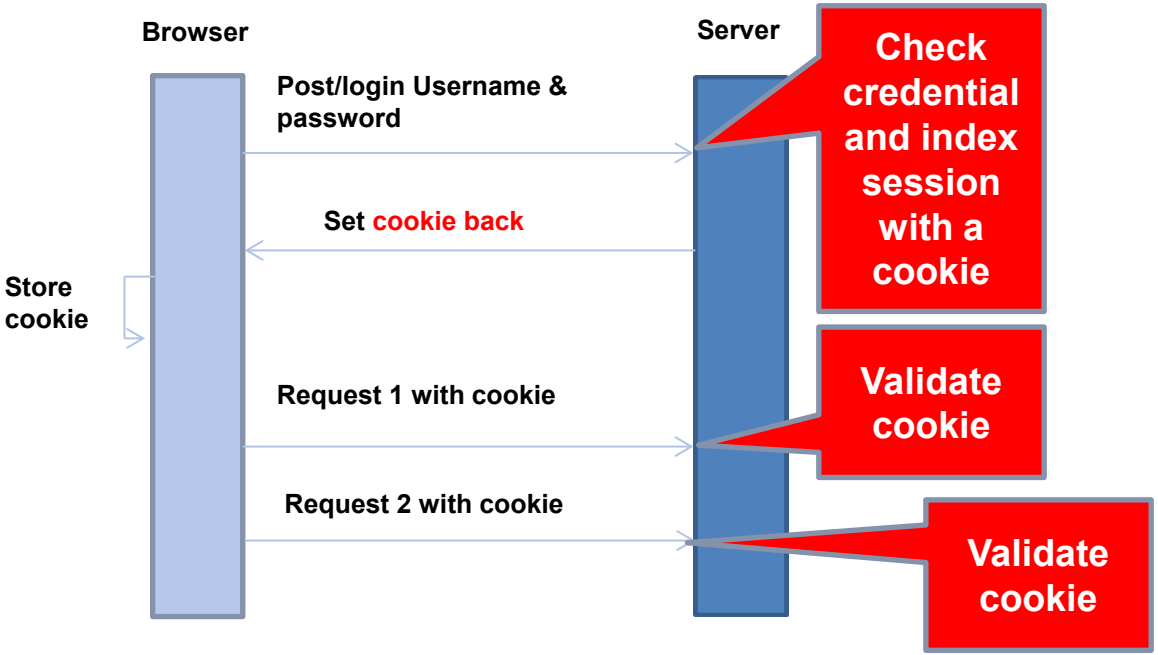
- In hidden form field

  <input type= “hidden” name = “sessionToken” value = “1234”>

- Browser cookie

  setcookie: sessionToken = 1234

Session management with cookie



How cookies work

- Setting and sending cookies
  - In header of HTTP response (Server to browser)

    set-Cookie: token=**1234**; expire=Wed, 3-Aug-2025 08:00:00; path=/; domain = idi.ntnu.no
  - In header of HTTP request (Browser to server, when visiting the domain of the same scope)

    Cookie: token=**1234**

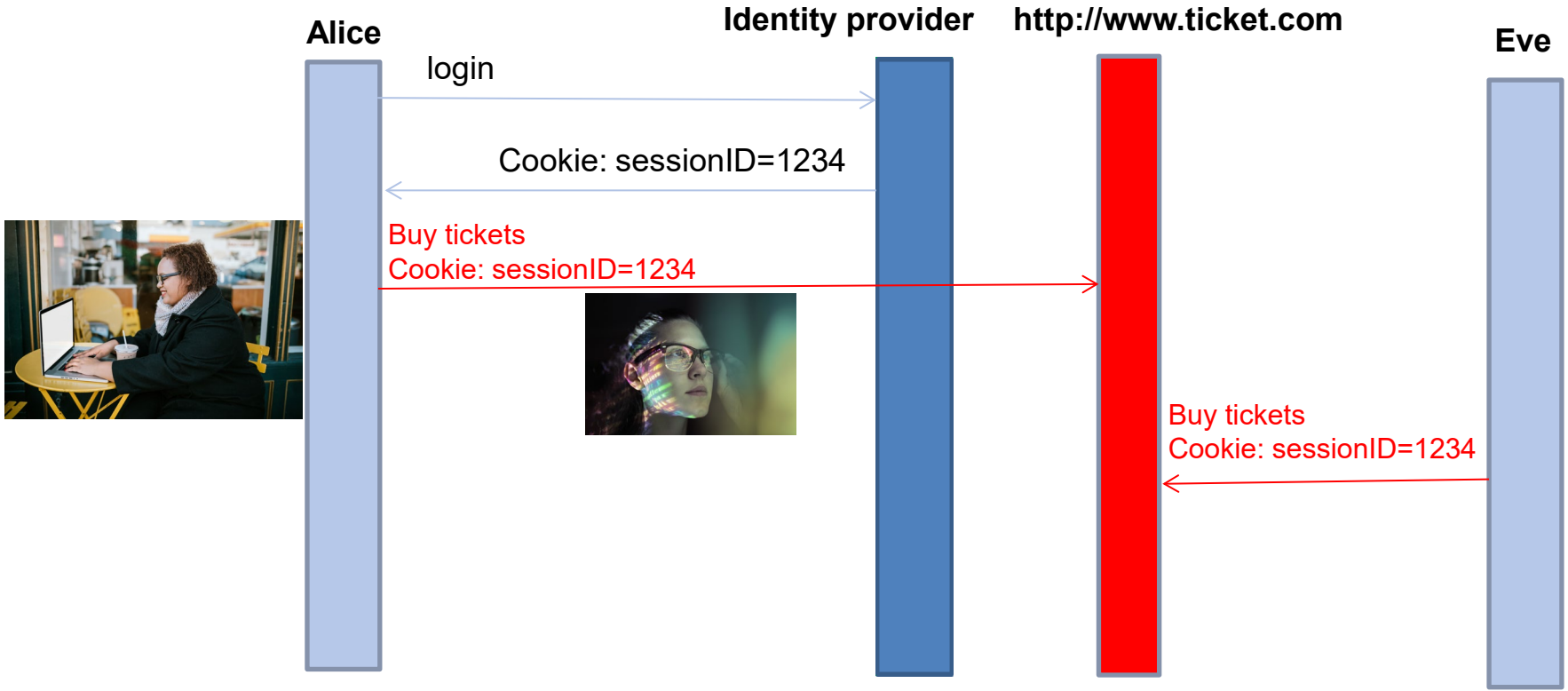
- Cookie protocol problem
  - Sever only sees Cookie: NAME = VALUE
  - Server does not see which domain sends the cookie

Vulnerable to session management attacks

Session management attacks and countermeasures

- Session token theft
- Session token predication attack
- Session fixation attack

Session token theft – Sniff network



Session token theft – Logout problem

- What should happen during logout
  - 1. Delete session token from the client
  - 2. Mark session token as expired on the server
  - Many do (1) but not (2)!!
- Attacker
  - If he can impersonate once, he can impersonate for a long time
  - E.g., Twitter sad story
    - Tokens not invalidated, replay attacks!

      https://packetstorm.news/files/id/119773

Solutions to Session token theft

- Once user logged in (i.e., session token issued), all later communication between browser and server shall use an encrypted channel (e.g., HTTPS)
- Remember to log out
- Time-out session ID
- Delete expired session ID
- Binding session token to the client's IP or computer

More about cookies

Session cookies

- Temporary cookies stored in the browser's memory just until the browser is closed
- Lower risks
- E.g. Online banks

Persistent cookies

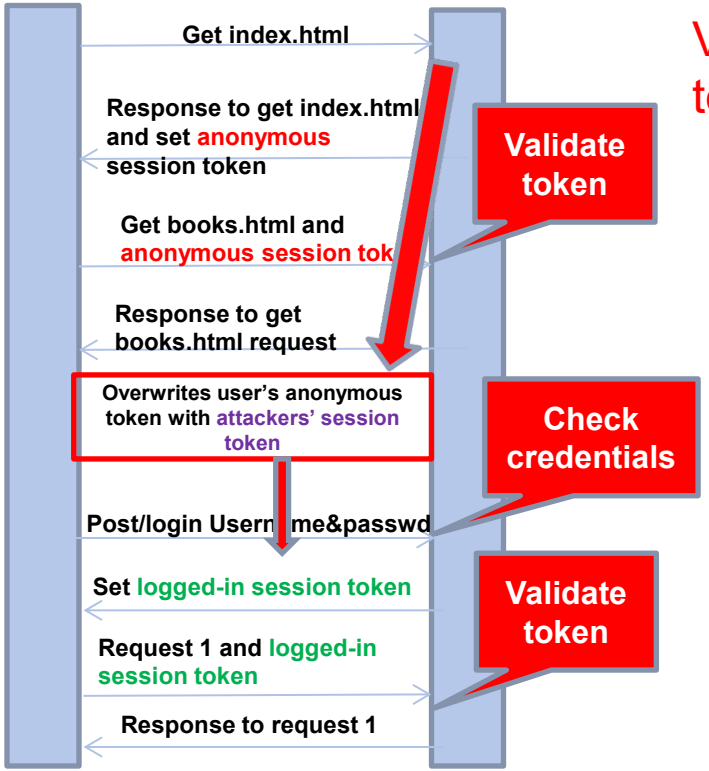
- Longer-term cookies that are tagged by the issuer with an expiration date
- Stored by the browser even after the browser is closed
- E.g., Google or Facebook to create a log of user activity
- When clicking "Remember me"

https://www.cookiebot.com/en/session-cookies/

Session token predication attack

- Predicable tokens, e.g., counter
  - jsessionid=user001
  - jsessionid=user002
  - jsessionid=user00?
- Non-predicable token: Seeing one or more token, should not be able to predict other tokens
- Solution:
  - Do not invent your own token generator algorithm
  - Use token generators from frameworks (e.g., ASP, Tomcat, Rails, Django)

Session fixation attack



Vulnerability: Server elevates the anonymous token without changing the value

Attack steps

1. User (e.g., Alice):
   Visits site using an anonymous token
2. Attacker
   Overwrites user's anonymous token with own token
3. User:
   Logs in and gets anonymous token elevated to logged-in token
4. Attacker:
   Attacker's token gets elevated to logged-in token after user logs in

How to overwrite session token?

- Tampering through network
  - Alice visits server using non-encrypted channel (HTTP)
  - The attacker injects into the response to overwrite the secure cookie

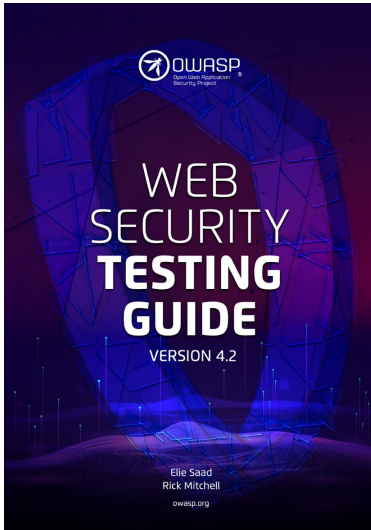
    Set-cookie: SSID=maliciousToken;
- Cross-site scripting (XSS)
  - How? Will explain more in XSS attack slides

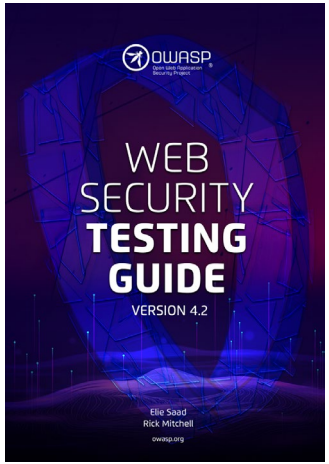
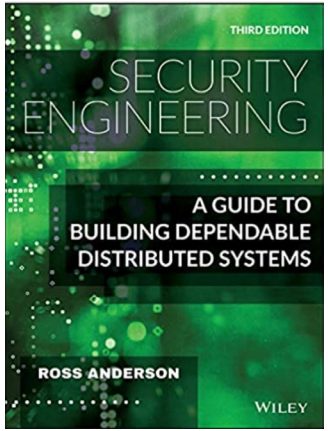
Mitigate session fixation

- Always issue a new session token, when elevating from anonymous token to logged-in token

Session management tests

- Testing for Bypassing Session Management Schema (WSTG-SESS-01)
- Testing for Cookies attributes (WSTG-SESS-02)
- Testing for Session Fixation (WSTG-SESS-03)
- Testing for Exposed Session Variables (WSTG-SESS-04)
- Testing for logout functionality (WSTG-SESS-06)
- Test Session Timeout (WSTG-SESS-07)
- Testing for Session puzzling (WSTG-SESS-08)



To read before next lecture

- OWASP Testing guide
  - Authentication testing
  - CSRF testing
  - XSS testing (Cross-site scripting)
  - SSRF testing (Server-side request forgery)
- Security engineering book
  - Chapter 3.4 Passwords
  - Chapter 3.5 CAPTCHAs